



POLITECNICO
MILANO 1863

INTRODUCTION TO SIMULATION

Dynamical Models and Current Control Methods

Dynamics of Electrical Machines and Drives (10CFU)

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Electrical Machines, Drives and Power Electronics Research Group

Contacts

Campus Bovisa

Electrical machines, Drives and power electronics research group

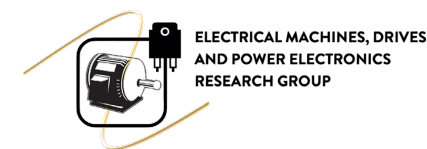
□ Location:

Department of Mechanical Engineering
Building B23, 1st floor
Office: Prof. Francesco Castelli Dezza
Tel. 8455

□ Appointments for questions & explanations: [send an email](#)

- **Ing. Matteo Sposito** - matteo1.sposito@polimi.it
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News: <https://www.linkedin.com/company/epebbs>



Practical Lectures Overview

Exercises

Practical lectures will take place at:

- ❑ [Aula T.2.2](#)
Building 13 (Trifoglio)
Campus Leonardo
Via Bonardi, 9 - 20133 - Milano (MI)

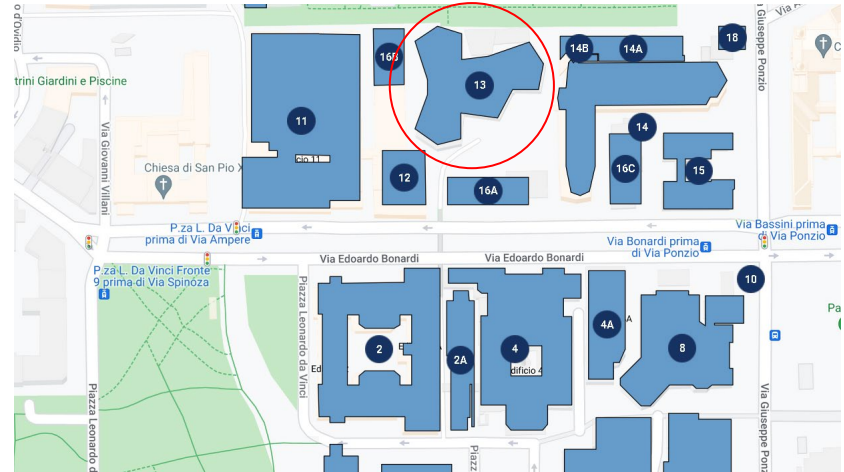
Tuesday 16:30 – 18:30

- ❑ course pages links:

[course webpage 1](#) [course webpage 2](#) [course webpage 3](#)

- ❑ Students are required to bring their own PCs

- ❑ Three *Lecturer* to support students both in-person and remotely



Ing. Matteo Sposito



Ing. Andrea Polastri



Ing. Mattia Rossi

Practical Lectures Overview

Exercises

❑ The aim of these lectures is to be **practical** (!)

❑ **Each lecture will follow this structure:**

- 1) a *first introductory part* to explain the lecture topic ($\approx$ 1h)
- 2) a *second part* in which one (or more) *exercise(s)* is (are) proposed to the students, who are going to do it (them) with their on PCs (coffee break)
($\approx$ 1h)
 - do the exercises in class is not mandatory, but recommended since you can ask questions to **Ing. Sposito** / **Ing. Polastri** who are available to support you (and provide you insights/suggestions)

Answering questions:



✓ **In-person**

Ing. Matteo Sposito

email: matteo1.sposito@polimi.it



✓ **In-person**

Ing. Andrea Polastri

email: andrea.polastri@polimi.it



✓ **For people connected remotely**

Ing. Mattia Rossi

email: mattia.rossi@polimi.it

Practical Lectures Overview

Exercises

- ❑ The exercises relates to using computer simulations tools (MATLAB/Simulink) to:
 - analyze the behavior of a dynamic system
 - analyze the effect of the control loop on the performance of an electrical drive

Final exam

The exam consists of an oral test which verify:

- the knowledge of the control of the electric drives (Prof. Castelli Dezza lectures)
- and a discussion on the report provided by the student (provide a report is mandatory)



during the practical lectures, the students will encounter both:

- **Standard exercises** (not suitable for the report)
- **Report exercises** (suitable for the report)
 - ✓ there will be four of this exercises during the semester
 - ✓ the students can pick one of them and prepare the report (at any time)
 - ✓ the report has to show the definition of the parameters of the control system and the results/figures of the Simulink simulation



NB exercises can be done as groups... BUT each student MUST present its own report

Systems Theory

Dynamical System

Definition

a **dynamical system** describes the evolution of a state over time

we need to specify what we mean for “evolution”, “state” and “time”:

① **continuous time** $t \in \mathbb{R}$

the evolution of the state is described by *ordinary differential equations* (ODE). Think of the linear, continuous time system in state space form

$$\dot{x}(t) = Ax(t)$$

② **discrete time** $k \in \mathbb{Z}$

the evolution of the state is described by a difference equation. Think of the linear discrete time system in state space form

$$x(k+1) = Ax(k)$$

Models for Continuous Systems

Dynamical Model

- A **dynamical model** of a system is a set of *mathematical laws* explaining in a compact form and in quantitative way how the system evolves over time

dynamical model $\Leftrightarrow$ mathematical model

$$v(t) = Ri(t) \quad i(t) = C \frac{dv(t)}{dt} \quad v(t) = L \frac{di(t)}{dt}$$

- Main questions about dynamical system and their model:
 - ① How to build a model (“How X and Y influence each other?”)
 - ② Simulation (“What happens if I apply action Z on the system?”)
 - ③ Design (“How to make the system behave the way I want?”)

Simulation of Dynamical Models

Conflicting Objectives

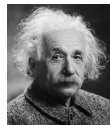
Experiments provide an answer, but have limitations:

- maybe too expensive (e.g.: launch a space shuttle)
- maybe too dangerous (e.g.: a nuclear plant)
- maybe impossible (the system doesn't exist yet!)

Simulating a dynamical model has zero-cost compared to real experiments
...but has conflicting objectives

- *Descriptive enough* to capture the main behavior of the system
- *Simple enough* for analyzing the system

“Make everything as simple as possible, but not simpler.” - *Albert Einstein*



A. Einstein
(1879-1955)

MATLAB/Simulink

an Equation Solver (ES)

we are going to use **MATLAB** as simulation environment
it is based on matrix algebra, developed from Prof. Cleve Moler
(for his students, as Fortran's interface in early '80s)



C. Moler
(founder)

- it's simple to implement control systems using **Simulink**

Good Things

- writing of mathematical equations
- library ready-to-use (e.g. SimPowerSystems)

Bad Things

- manage computational error (e.g. numerical integration)
- simulation time t_s could differ from real time t (e.g. $t_s = 10s \leftrightarrow t = 1h$)



use [MATLAB 2018b version](http://software.polimi.it) (<http://software.polimi.it>)

Basics for Electrical Drives Simulation

Current Control of an RL load

- electrical machines windings behaves as **RL (ohmic-inductive) load**
- controlling the *current flow* into windings imply *control the torque*

Kirchhoff's voltage law :
$$v(t) - L \frac{di(t)}{dt} - Ri(t) = 0$$

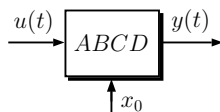
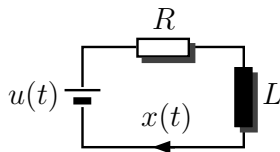
- 1 Rewrite it as a 1st order linear system

$$\frac{di(t)}{dt} = \frac{1}{L}v(t) - \frac{R}{L}i(t)$$

- 1 or in a **state-space form**

$$\dot{x}(t) = \underbrace{-R/L}_{A} x(t) + \underbrace{1/L}_{B} u(t)$$

where $x(t) = i(t)$ and $u(t) = v(t)$



Basics for Electrical Drives Simulation

Evolution of the State (1st order LTI)

- ① We want to **observe** and control the currents $x(t) = i(t) \rightarrow y(t) = x(t)$
- ② Voltage $u(t) = v(t)$ is a **forcing signal**
- ③ Adding the information about **initial conditions** $x_0 \in \mathbb{R}^n$

$$\begin{cases} \dot{x}(t) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t) + Du(t) \\ x(0) &= x_0 \end{cases}$$

where

$$A = -\frac{R}{L} \quad B = \frac{1}{L} \quad C = 1 \quad D = 0$$

- the **evolution of the state** $x(t) = y(t)$ is

$$x(t) = \underbrace{e^{At} x_0}_{\substack{\text{natural response} \\ \text{(effect of initial condition)}}} + \underbrace{\int_0^t e^{A(t-\tau)} B u(\tau) d\tau}_{\substack{\text{forced response} \\ \text{(effect of input signal)}}$$

- if $A < 0 \Rightarrow$ **asymptotically stable**

Basics for Electrical Drives Simulation

Transfer Functions

- **Laplace transforms** convert integral and differential equations into algebraic equations through the *Laplace operator*

$$F(s) = \mathcal{L}[f(t)] \quad \leftarrow \quad s = d/dt$$

- It can be demonstrated that

$$y(s) = \underbrace{C(sI - A)^{-1}x_0}_{\text{Laplace transform of natural response}} + \underbrace{(C(sI - A)^{-1}B + D)u(s)}_{\text{Laplace transform of forced response}}$$



P.S. Laplace
(1749-1827)

Definition

The **transfer function** $G(s)$ of a continuous-time linear system (A, B, C, D) is

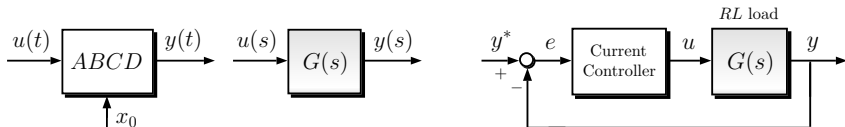
$$G(s) = C(sI - A)^{-1}B + D$$

between the Laplace transform $y(s)$ of output and the Laplace transform $u(s)$ of the input signals for the *initial state* $x_0 = 0$

Basics for Electrical Drives Simulation

Current Control (PI-based) of an RL Load

Note that $y(s) = i(s)$ and $u(s) = v(s)$



RL load - $G(s)$

$$G(s) = \frac{y(s)}{u(s)} = \frac{1}{R + sL}$$

Current Controller - $R(s)$

- P, PI, PID ☒
- Optimal Control

- The *RL* load to be controlled has a time constant $T_G = L/R$
- The controller settings are chosen according to:
 - ① **Time response** $\rightarrow$ **Bandwidth** ω_c
 - ② **Robustness level** $\rightarrow$ **Phase margin** φ_m
 - ③ **Sensitivity to disturbances (and actuations)** $\rightarrow S(s)$

P-Action

Minimum Phase Systems

- Consider a P controller $R(s) \rightarrow u(s) = k_p e(s)$
- The design will be based on the **open-, closed-loop** transfer functions

$$L(s) = R(s)G(s) = \frac{k_p}{R + sL} \quad F(s) = \frac{L(s)}{1 + L(s)} = \frac{k_p}{k_p + R + sL}$$

but $L(s) = u(s)/e(s)$ is included in the category of:

Minimum Phase Systems (mps)

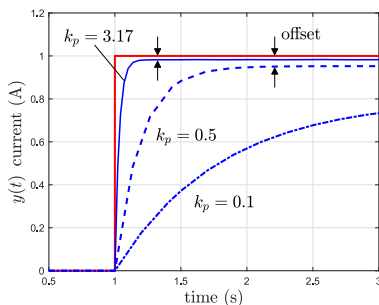
If an LTI system presents: *positive gain, poles $\Re(p) < 0$, zeros $\Re(z) < 0$* it is a **minimum phase system**

- Then, it is enough an $L(s)$ cutting the 0dB axis with a slope -20dB/decade to guarantee $\varphi_m \cong 90^\circ$ and asymptotic stability
- Bandwidth is $|L(j\omega_c)| = 1 \Rightarrow \omega_c = \sqrt{k_p^2 - R^2}/L$

Property of P-Action

Step Response

- **Data:** $R = 0.025$ and $L = 0.1H \rightarrow \tau_G = 4s$ and $T_A^{(G)} = 5\tau_G = 20s$
- **Desired characteristics:** $T_A^{(F)} = 1s \rightarrow \tau_F = T_A^{(F)} / 5 = 0.2s$



- 1 dominant pole

$$\omega_c = 2\pi/\tau_F = 31.4 \text{ rad/s}$$

- 2 proportional gain

$$k_p = \sqrt{R^2 + \omega_c^2 L^2} = 3.17 \text{ } (\Omega)$$

increasing k_p the reference-tracking is better but we still have an offset (it could be 0 only if $k_p \rightarrow \infty$)

- The offset is computed through the **final value theorem**

$$(\text{f.v.t.}) \quad \lim_{s \rightarrow 0} s F(s) \frac{1}{s} = \lim_{s \rightarrow 0} \frac{k_p}{k_p + R + sL} = \frac{k_p}{k_p + R}$$

PI-Action #1

NO - Pole/Zero Cancellation

Skip

- Consider a PI controller $R(s) \rightarrow u(s) = (k_p + k_i/s) e(s)$

$$L(s) = \frac{sk_p + k_i}{s(R + sL)} \quad F(s) = \frac{sk_p + k_i}{k_i + s(k_p + R) + s^2L}$$

- Exploit the relationship between $L(s)$ and $F(s)$

$$|F(s)| = \frac{|L(s)|}{|1 + L(s)|} \begin{cases} 1 & \forall \omega : |L(s)| \gg 1 \rightarrow \omega \ll \omega_c \\ |L(s)| & \forall \omega : |L(s)| \ll 1 \rightarrow \omega \gg \omega_c \end{cases}$$

- It can be demonstrated

$$|F(j\omega_c)|_{s=j\omega_c} = \frac{|L(j\omega_c)|}{|1 + L(j\omega_c)|} = \frac{1}{2 \sin(\varphi_m/2)} \quad \begin{matrix} \varphi_m \cong 90^\circ \\ \rightarrow \\ (\text{mps}) \end{matrix} \quad |F(j\omega_c)|^2 = \frac{1}{2}$$

explicit characteristic equation from $F(j\omega)$

$$|F(j\omega_c)|^2 = \frac{1}{2} \rightarrow k_i^2 + 2\omega_c^2 k_i L + \omega_c^2 k_p^2 - \omega_c^2 (R^2 + 2k_p R) - \omega_c^4 L^2 = 0$$

PI-Action #1

NO - Pole/Zero Cancellation

Skip

- ① The solution of the 2nd order eq. leads to

$$k_i = -\omega_c^2 L \pm \underbrace{\omega_c \sqrt{2\omega_c^2 L^2 - k_p^2 + R^2 + 2k_p R}}_{>0}$$

- ② Note that k_i must be $k_i > 0 \cap k_i \in \mathbb{R}$, then

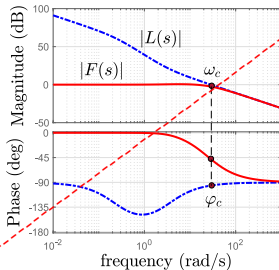
$$R - \sqrt{2R^2 + \omega_c L^2} < k_p < \underbrace{R + \sqrt{2R^2 + \omega_c L^2}}_{k_{p,\max}}$$

$$k_{p,\max} = R + \sqrt{2R^2 + \omega_c L^2}$$

- a practical rule is $k_p = 0.9k_{p,\max}$

$$\varphi_m = \pi - |\angle L(j\omega_c)| \rightarrow 90^\circ$$

$$\angle L(j\omega_c) = \frac{\pi}{2} - \operatorname{atan}\left(\frac{k_i}{\omega_c k_p}\right) - \operatorname{atan}\left(\frac{L}{R}\right)$$



PI-Action #2

YES - Pole/Zero Cancellation

$$k_p = w^* L \rightarrow k_p / k_i = L / R$$

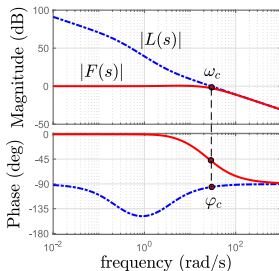
$$k_i = w c^* R$$

$$k_p = w c^* R^* T_i = w c^* R^* L / R = w c^* L$$

- An alternative form $R(s) = k_p (1 + sT_i) / sT_i$ where $T_i = k_p / k_i$
- Lets make a step-back and rewrite the expression of $L(s)$:

$$L(s) = R(s)G(s) = k_p \frac{1 + sT_i}{sT_i} \frac{1}{R_s + sL} = \frac{k_p}{R_s} \frac{\cancel{1 + sT_i}}{sT_i} \frac{1}{\cancel{1 + s(L/R_s)}}$$

- It is possible to make **pole/zero cancellation** by $T_i = L / R_s$



$$L(s) \Rightarrow \frac{k_p}{R_s s T_i}$$

- as already did:

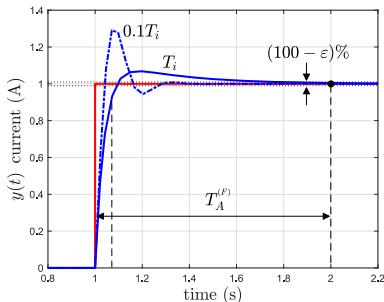
$$|L(s)|_{s=j\omega_c} = 1 \rightarrow \omega_c = \frac{k_p}{R_s T_i}$$

- different behaviour but still [mps](#)

Property of PI-Action

YES - Pole/Zero Cancellation

- Considering the two ways of control design we have:
- Desired characteristics:** $T_A^{(F)} = 1s \rightarrow \tau_F = T_A^{(F)} / 5 = 0.2s$



- proportional/integral gains

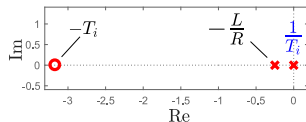
$$k_p = 3.17 (\Omega) \quad k_i = 9.03 (\Omega s)$$

- dominant pole

$$\omega_c = 2\pi/\tau_F = 31.4 \text{ rad/s}$$

that is related to T_i as

$$\Rightarrow T_i = k_p / R_s \omega_c$$

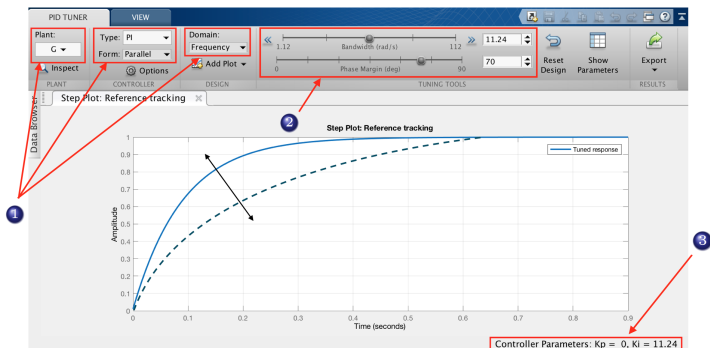


- higher is the integral action, faster is the control but with an oscillating response

PI-Action #3

Direct Approach with MATLAB - pidtool®

- Given the plant $G(s)$ this interface allow you to select the controller type (P,PI,PID) and directly see the effects of parameter changing



- Hint:**

- *frequency criterium* (in order to use your Bode knowledge)
- needs to know *natural dynamics* as before

Exercise

Current Control of an RL load

Given the previous RL load, test the following current control action:

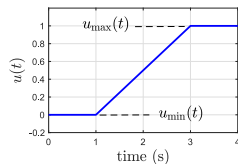
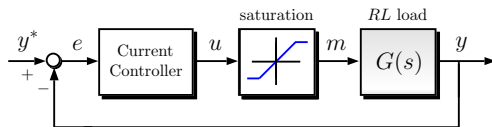
- | | |
|----------------|---------------------------------|
| ① P-Action | $R=0.025 \quad L=0.1$ |
| ② PI-Action #2 | $Ta_f=1 \quad \tau_f=0.2$ |
| ③ PI-Action #3 | $wc = 1/\tau_f \text{ (rad/s)}$ |

Hint:

- Write a MATLAB script to compute transfer functions $G(s)$, $L(s)$, and $F(s)$
- Test different ω_c to verify the previous choice
- Try, make mistakes....

Nonlinear Effects

Actuator Saturation



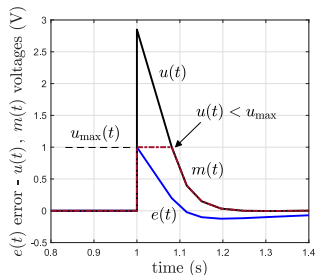
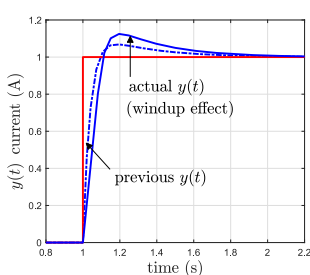
- A linear controller is simple to design, but the performances are good as long as the dynamics remain *close* to linear theory
- Nonlinear effects require care, such as **actuator saturation**
 - Saturation can be defined as the **static nonlinearity**

$$\text{sat}(u(t)) \begin{cases} u_{\min}(t) & \text{if } u(t) < u_{\min}(t) \\ u(t) & \text{if } u_{\min}(t) < u(t) < u_{\max}(t) \\ u_{\max}(t) & \text{if } u(t) > u_{\max}(t) \end{cases}$$

$u_{\min}(t)$ and $u_{\max}(t)$ are min and max allowed actuation signals (e.g. $\pm 1\text{ V}$ for a voltage supply)

The Windup Problem

Integrating Error (Windup)

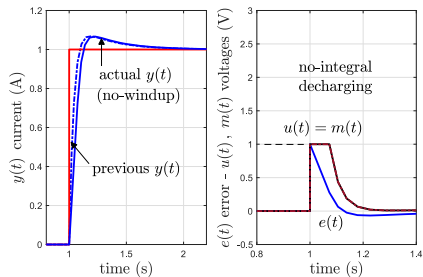
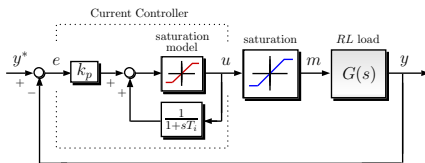


- The output $y(t)$ takes a longer time to achieve steady-state due to “windup” of the integrator (higher peak response)
- If $e(t)$ increase, $u(t)$ increase even if $m(t)$ is saturated to u_{max} , the PI **keeps integrating the tracking error** $e(s) \rightarrow$ producing $u(t) \neq m(t)$
- If $e(t)$ change sign we have to wait until $u(t) < u_{max}$ before to come back into linear region $m(t) = u(t)$, this means wait for *integral discharge*

Anti-Windup Techniques

Saturation Model

- **Practical issue?** A controller without any information about saturation
- There are several techniques that have in common the idea to augment the controller with *saturation information* (or dynamical model)



- This scheme has no effect when the actuator is not saturating, while keeps same behavior between $u(t)$, $m(t)$ when $e(t)$ change sign
- Integrator windup is avoided thanks to back-info

Exercise

Current Control of an RL load

An RL load, $R = 1\Omega$ and $L = 1mH$, is fed by a voltage supply ($\pm 30V$). Design a PI controller in order to follow:

- ① a step-command from 0 to 10A in 1s
- ② a sinusoidal-command of 10A at 5Hz

Hint:

- Write a MATLAB script to compute k_p and k_i given R , L , and ω_c
- Test different ω_c to verify the previous choice